

Dr. Lenard Dome

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RESEARCH AREAS

Computational Experimentation

Computational Modelling of Learning and Categorisation

ACADEMIC POSITIONS

2023 - Postdoctoral researcher. University of Tuebingen. (Advisor: Tobias Hauser)
2023 - Guest Scientist. Max Planck Institute for Biological Cybernetics.
2020 - 2023 Ph.D. Psychology. Plymouth University. (Supervisor: Andy Wills).
2021 - 2023 Visiting Lecturer, Tutor and Specialist. University of Plymouth.
2019 - 2020 M.Sc. Advanced Psychology. Plymouth University. (Supervisor: Patric Bach).
2019 - 2023 Teaching Assistant. University of Plymouth.
2017 - 2019 Research Assistant. (Line Manager: Andy Wills).
2017 - 2018 Research Assistant. (Line Manager: Marina Wimmer).

EDUCATION

2020 - 2023 Ph.D. Psychology. Computational Cognitive Science. *Straight Pass*.
2019 - 2020 M.Sc. Advanced Psychology. Plymouth University. *Distinction*.
2017 - 2019 B.Sc. Psychology. Plymouth University. *First*.

THESIS

Dome, L. (2023) *Testing the adequacy of formal models of an irrational learning effect*. University of Plymouth, UK.
Defended: 2023.09.12. <https://pearl.plymouth.ac.uk/psy-theses/74>

GRANTS AND AWARDS

2020 **Awarded** [Undergraduate Project Prize](#). Experimental Psychological Society / British Science Association.
2019 **Awarded** Doctoral Training Studentship (1+3). ESRC & SWDTP. £87,510.62.

PUBLICATIONS

- in.prep **Dome, L.**, Moses-Payne, M. E., Hauser, T. U., “Age-related shifts in metacognition reveal converging confidence for correct and error judgments across the lifespan.”
- 2026 Schlegelmilch, R. & **Dome, L.** Snake oil or Panacea? How to Misuse AI in Scientific Inquiries of the Human Mind. *Behavioural Sciences*. <https://doi.org/10.3390/bs16020219>.
- 2025 Marzuki, A., Kosina, L., **Dome, L.**, Hewitt, S., & Hauser, T. Metacognitive antecedents to states of mental ill-health: Drops in confidence precede symptoms of OCD. *Research Square*. <https://doi.org/10.21203/rs.3.rs-7544256/v1>.
- 2025 **Dome, L.**, Kadri, K., Wagner, B. J., Webb, A. & Hauser, T. U. “cpm: a python library for theory-driven modeling in Computational Psychiatry.” *PsyArxiv*. https://doi.org/10.31234/osf.io/6jc3t_v1.
- 2025 **Dome, L.** & Wills, A.J. “g-Distance: On the comparison of model and human heterogeneity.” *Psychological Review*. <https://doi.org/10.1037/rev0000550>
- 2025 **Dome, L.** & Wills, A.J. “Better generalization through distraction? Concurrent load reduces the size of the inverse base-rate effect.” *Psychonomic Bulletin & Review*. <https://doi.org/10.3758/s13423-025-02661-1>
- 2023 **Dome, L.** & Wills, A. J. “Errorless irrationality: removing error-driven components from the inverse base-rate effect paradigm.” *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. <https://escholarship.org/uc/item/0kw671vv>
- 2021 **Dome, L.**, Edmunds, C. E. R., Wills, A. J. “SUSTAIN captures category learning, recognition, and hippocampal activation in a unidimensional vs information-integration task.” *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. <https://escholarship.org/uc/item/5r98q3dr>
- Spicer, S. G., Wills, A. J., Jones, P. M., Mitchell, C., & **Dome, L.** “Representing uncertainty in the Rescorla-Wagner model: blocking, the redundancy effect, and outcome base rate.” *Open Journal of Experimental Psychology and Neuroscience*. doi.org/10.46221/ojepn.2021.6623
- 2019 Wimmer, M. C., **Dome, L.**, Wennekers, T. & Hancock, P. J. B. “Is the letter cancellation task a suitable index of ego-depletion? Empirical and conceptual issues.” *Social Psychology*. doi.org/10.1027/1864-9335/a000393

RESEARCH SOFTWARE

-  **psp** *maintainer, developer* Implements an n -dimensional parameter space partitioning algorithm for evaluating the global behaviour of formal computational models as described by Pitt, Kim, Navarro and Myung (2006). 11K downloads. [\[source code\]](#) [\[cran\]](#) doi: [10.32614/CRAN.package.psp](https://doi.org/10.32614/CRAN.package.psp)
-  **cpm** *maintainer, developer* cpm is a fundamental python package for Computational Psychiatry. It is designed to provide a set of tools for researchers and clinicians to analyze and model data in the field of Computational Psychiatry. [\[source code\]](#)
-  **catlearn** *senior developer* Catlearn is an archive of formal models of categorization and learning, plus benchmark datasets to test them against. It currently houses 14 formal mathematical models of learning and categorization. I have contributed 8 models to the package. 45K downloads. [\[source code\]](#) [\[cran\]](#) doi: [10.32614/CRAN.package.catlearn](https://doi.org/10.32614/CRAN.package.catlearn)

INVITED TALKS

- 2026 **Yale University (USA)**
MAP Talk. *cpm*: a Python library for theory-driven modeling in computational psychiatry.
- 2024 **University of Tübingen (Germany)**
Psychiatry Colloquium. *cpm*: a Python library for theory-driven modeling in computational psychiatry.
- 2021 **British Science Association / Experimental Psychological Society (UK)**
Undergraduate Project Prize Talk (2020). Clearing confounds from the inverse base-rate effect: irrationality and concurrent load. [Video](#).
- Liverpool John Moores University (UK)**
Cognitive and Affective Neuroscience Group Seminar. The role of base rates in categorizing ambiguity and forming stereotypes.

CONFERENCE PRESENTATIONS

- 2025 Computational Psychiatry Conference. Tuebingen, Germany.
- 2023 45th Annual Conference of the Cognitive Science Society. Sydney, Australia.
- 2023 MathPsych/ICCM/EMPG: Amsterdam, The Netherlands.
- 2023 Meeting of the Experimental Psychology Society. Plymouth.
- 2022 22nd Associative Learning Symposium: Gregynog Hall, Wales, UK.
- 2021 Cognitive Science Conference: Online.
- 2019 The British Psychological Society: South West Undergraduate Conference. Plymouth, UK.

SUPERVISION AND TEACHING

- 2025 **University of Tübingen (Germany)** Workshop: Introduction to computational modeling (M.Sc., Ph.D., Postdocs).
- 2025 **Max Planck Institute for Biological Cybernetics (Germany)** [CaCTüS](#) internship student supervision (B.Sc.).
- 2024–2025 **University of Tübingen (Germany)** Seminar: Introduction to computational modeling (B.Sc., M.Sc.).
- 2022–2023 **University of Plymouth (UK)** Workshop: Introduction to quantitative analysis with R (DCLinPsy).
- 2021–2023 **University of Plymouth (UK)** Final-year project supervision (B.Sc.).
- 2021–2023 **University of Plymouth (UK)** Seminar: Bayesian sample size estimation // Data visualization in R (B.Sc.).
- 2021–2023 **University of Plymouth (UK)** Research methods in R (B.Sc.).

SOCIETY MEMBERSHIPS

Cognitive Science Society Member
Experimental Psychological Society

German Centrum for Mental Health (DZPG) ECS
European Brain and Behaviour Society

PEER-REVIEW

The American Statistician.